Shirlene Karimi
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SOFTWARE CAT 3

MODULE 3,TERM 3

Instructions

Answer all questions

QUESTIONS

1. Define the term **API** and give one real-life example of its use. (2 marks)
2. State **three** differences between REST and SOAP. (3 marks)
3. Define an **API endpoint** and give one example. (2 marks)
4. Outline **four** main features of Postman. (4 marks)
5. State the HTTP method used to perform each of the following actions: (4 marks)
   (a) View student records
   (b) Add a new student
   (c) Edit a student's details
   (d) Remove a student record
6. State the meaning of each of the following HTTP status codes: (4 marks)
   (a) 201 (b) 401 (c) 404 (d) 500
7. Describe the steps you would follow in Postman to retrieve a list of users from https://jsonplaceholder.typicode.com/users using a GET request. (4 marks)
8. Define **corrective measures** and distinguish them from preventive and detective measures. (3 marks)
9. Name **four** software corrective tools and state the purpose of each. (4 marks)
10. Outline **five** steps followed when performing corrective measures on a faulty computer system. (5 marks)
11. Define **software testing** and state **two** of its objectives. (3 marks)
12. Differentiate between **smoke testing** and **regression testing**. (2 marks)
13. List **four** levels of software testing and briefly describe each. (4 marks)
14. Name the **four** types of software maintenance and give one example of each. (4 marks)
15. Define a **software maintenance schedule** and state **two** of its benefits. (2 marks)

1. API (Application Programming Interface) is a set of rules and protocols that allows two software applications to communicate and exchange data with each other.

Example.

When you use M-Pesa to pay for goods through an online shopping app, an API allows the shopping app to communicate with M-Pesa to process the payment and confirm whether it was successful.

2. REST uses JSON for data exchange while SOAP uses XML for message formatting.

REST is generally lightweight and faster while SOAP has more overhead due to its strict message structure.

REST commonly uses HTTP method while SOAP can use different transport protocols

3. An API endpoint is a specific URL where an application can access a particular resource.

Example

https://jsonplaceholder.typicode.com/users

4. Features of Postman

API testing

Request methods

Response viewing

Collections.

5. HTTP method used to perform:

i) View students records - GET

ii) Add a new student - POST

iii) Edit a students details - PUT

iv) Remove a students record - DELETE

6. HTTP Status code.

a) 201 - Created.

b) 401 - Unauthorized.

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7. i) Open the Postman application.
ii) Select GET from the request method dropdown.
iii) Enter the URL https://jsonplaceholder.typicode.com/users.
iv) Click Send and view the list of users in the response body.

8. Corrective measures are actions taken to fix a problem after it has occured.
Preventive measures are actions taken to stop problems from occurring.
Detective measures are actions taken to identify or detect problems when they occur.

9. Software corrective tools.
Antivirus software - it detects and removes malware infections.
Disk Repair Tool (CHKDSK) - it detects and repairs file-systems on a disk.
System Restore - it returns the computer to an earlier working state.
Debugger - it helps to locate and fix errors in a program.

10. Identify the problem.
Diagnase the cause of the problem.
Choose the appropriate corrective action.
Implement the corrective action.
Test the system to confirm that the problem has been resolved.

11. Software testing is the process of checking software to identify defects and verify that it meets specified requirements.

Objectives
To identify and correct software defects.
To ensure that the software meets user ~~requirm~~ requirements functions correctly.

12. Smoke testing is a preliminary test performed to check whether main functions of a new software build work.
Regression testing - the test is performed after changes to that existing functions

12. Smoke testing is a preliminary test performed to check whether the main functions of a new software build work while regression testing the testing is performed after changes to ensure that existing functions still work correctly.

13. Unit testing - it tests individual components or units of software
Intergration testing - it tests whether combined modules work correctly together.
System testing - it tests the complete software system.
Acceptance testing - it checks whether the software meets user requirements and is ready for use.

14. Corrective maintenance - fixing an error in software
Adaptive maintenance - Updating software to work with a new operating system.
Perfective maintenance - Improving software performance
Preventive maintenance - Modifying software to reduce the risk of future problems.

15. A software maintenance schedule is a planned timetable showing when software maintenance activities should be perf

Benefits
It helps prevent unexpected software failures.
It improves software reliability and performance.